CASH TO COMPANY

Item Bank

"*Know your customer*"

Per-unit quiz / test items · 4 units · ETS-quality alignment to Jump$tart, CEE, CCSS, C3, CASEL, NGSS-

Grade 7 — Cash to Company Item Bank

Year theme: *Know your customer.* This bank covers all four Grade 7 units (Who am I selling to? · Math of one customer · Honest stories that sell · Feedback that fits). Items use customer/persona/contribution-margin scenario style and proportional reasoning (CCSS 7.RP.A.3). Distractors target the Grade 7 Common Misconceptions table. No family-money probes; "for a hypothetical" framing throughout. Cash's Grade 7 voice: "*Revenue is loud. Contribution margin is true. Which people, and how do you know?*"

Unit 1: Who am I selling to?

MC1. *"For a hypothetical venture, your 'persona' is..."*

A. A demographic profile (12-year-old female) · B. A specific person built from real interview evidence (12-year-old who plays soccer Tues/Thurs and forgets breakfast on game days) · C. Your best guess about a typical customer · D. The first person who said they'd buy

Answer: B · **Standard:** *NGSS-ETS MS-ETS1-1 — Define the criteria and constraints of a design problem with sufficient precision.* · **Strand outcome (CBM, 7):** *I can build a one-page "customer persona" with a name, demographic detail, daily routine, top 3 frustrations, and 3 channels where I could reach them.* · **Misconception targeted:** "*Demographics ARE the persona*" — corrects directly.

MC2. *"A real customer interview is contaminated when..."*

A. The interview is short · B. You describe your idea before they finish describing their problem · C. You ask follow-up questions · D. The customer is younger than you

Answer: B · **Standard:** *CCSS-ELA W.7.7 — Conduct short research projects.* · **Strand outcome (PSIG, 7):** *I can draft a problem statement in the form "[Customer] struggles with [problem] when [context], because [cause]" using interview evidence, not assumptions.* · **Misconception targeted:** "*Customer interviews are sales pitches*" — previews Grade 9 Mom Test rule.

MC3. *"Which problem statement is strongest?"*

A. Soccer players are tired · B. People want healthy snacks · C. 7th-grade soccer players who skip breakfast on Tues/Thurs game days struggle to focus during 1st period because their last meal was 14 hours earlier · D. Schools should sell more food

Answer: C · **Standard:** *CCSS-ELA W.7.7.* · **Strand outcome (PSIG, 7):** *I can draft a problem statement in the form "[Customer] struggles with [problem] when [context], because [cause].*" · **Misconception targeted:** "*Vague problem statements are fine*" — corrects.

SA1. *"Build a 1-page persona for a hypothetical venture. Include: name, demographic detail (1), daily routine (1-2 sentences), top 3 frustrations, 3 channels to reach them."*

Model answer: "*Name: 'Sam.' Demographic: 12yo, plays Tues/Thurs travel soccer. Routine: wakes 6:45, no breakfast on game days, school 8-3, soccer 4-6. Frustrations: (1) hungry by 10am; (2) cafeteria options are slow lines; (3) parents' protein bars taste 'chalky.' Channels: Instagram DMs, the team group chat, the locker hallway between 1st and 2nd period.*"

Scoring guide (3-2-1): - **3:** All 5 elements specific; behaviors > demographics; channels are real, not generic. - **2:** 4 of 5 specific. - **1:** Mostly demographic.

SA2. *"Write a problem statement in the form '[Customer] struggles with [problem] when [context], because [cause]' using your persona's evidence."*

Model answer: "*Sam (12yo soccer player) struggles with focus and energy in 1st period on game days, because the school day starts at 8 with no breakfast eaten since 7pm the night before.*"

Scoring guide (3-2-1): - **3:** All 4 slots filled with specifics tied to interview evidence. - **2:** 3 of 4 slots specific. - **1:** Slots vague or filled with assumptions.

Performance Task: Persona Pack (90 min over 2 sessions; 3 real interviews required, recorded surrogates allowed if access is limited).

Child conducts 3 real customer interviews (or watches 3 recorded surrogates), builds a 1-page persona with the 5 elements, and writes a problem statement using interview evidence. **Rubric:** Customer Research Rubric (Grade 7): (1) 3 sources of evidence cited, (2) persona contains 3+ behaviors, (3) problem statement uses interview language not assumptions. **Strand outcome:** CBM + PSIG.

Extension (advanced learner):

"*After your interviews, list 2 things you BELIEVED on Day 1 that the evidence changed. Quote the customer.*" Targets the Day-1-vs-Day-30 revision move; previews the year's LRCI Capstone outcome.

Unit 2: Math of one customer

MC1. *"For a hypothetical bracelet stand: $4 per bracelet revenue, $1.20 materials, $0.50 packaging. Contribution margin per bracelet?"*

A. $4.00 · B. $2.30 · C. $5.70 · D. $1.70

Answer: B · **Standard:** *CCSS-Math 7.RP.A.3 — Use proportional relationships to solve multistep ratio and percent problems.* · **Strand outcome (MFL, 7):** *I can calculate per-customer revenue, cost of acquisition (proxy), and contribution margin for a small business scenario, and explain why all three matter.* · **Misconception targeted:** "*Higher revenue = better business*" — A baits revenue-as-CM; B = $4 - $1.20 - $0.50 = $2.30.

MC2. *"Stand A: $20 revenue/customer, $18 variable cost/customer. Stand B: $5 revenue/customer, $1.50 variable cost/customer. Which has stronger UNIT economics?"*

A. Stand A (higher revenue) · B. Stand B (CM = $3.50 vs. $2; better margin) · C. Tied · D. Can't tell without volume

Answer: B · **Standard:** *CCSS-Math 7.RP.A.3.* · **Strand outcome (MFL, 7):** *I can calculate per-customer revenue, cost of acquisition (proxy), and contribution margin for a small business scenario.* · **Misconception targeted:** "*Higher revenue = better business*" — corrects directly with a Stand-A-vs-Stand-B example pulled from the Lesson Plans.

MC3. *"You spent $30 on flyers and got 12 new customers. Cost of acquisition (proxy)?"*

A. $30 · B. $2.50 · C. $360 · D. $0.40

Answer: B · **Standard:** *CCSS-Math 7.RP.A.3.* · **Strand outcome (MFL, 7):** *I can calculate per-customer revenue, cost of acquisition (proxy), and contribution margin.* · **Misconception targeted:** "*Marketing cost doesn't count*" — $30 / 12 = $2.50 per acquired customer.

SA1. *"For a hypothetical venture: $7 sale price, $2.50 materials, $0.75 delivery, $1 marketing-spend per customer. Compute revenue, contribution margin, and margin %."*

Model answer: "*Revenue/customer = $7. Variable cost/customer = $2.50 + $0.75 + $1.00 = $4.25. CM = $7 - $4.25 = $2.75. CM % = $2.75 / $7 = 39%.*"

Scoring guide (3-2-1): - **3:** All three numbers correct AND % is rounded sensibly. - **2:** CM correct, % off. - **1:** CM wrong.

SA2. *"Why do per-customer revenue, contribution margin, AND CAC ALL matter together? Answer in 2-3 sentences."*

Model answer: "*Revenue per customer tells you how much each sale is worth. Contribution margin tells you how much of that revenue is actually yours after the cost of serving them. CAC tells you what it cost to win them in the first place. If CAC > CM, you lose money on every customer no matter how many you have.*"

Scoring guide (3-2-1): - **3:** All three terms correctly differentiated AND the "CAC > CM = loss" relationship is named. - **2:** Two of three differentiated. - **1:** Mostly tautological.

Performance Task: Contribution Margin Worksheet + Pricing Recommendation (90 min).

Child computes per-customer revenue, variable cost, CM, CM%, and CAC for their venture, then writes a 1-paragraph pricing recommendation. **Rubric:** Budget Project Rubric (Grade 7): (1) all unit-economics numbers correct, (2) recommendation grounded in numbers AND customer willingness, (3) recommendation acknowledges one risk. **Strand outcome:** MFL + CBM.

Extension (advanced learner):

"*Your CM is $2.75; your CAC is $1. Your competitor's CM is $4 with CAC of $3.50. Which business is more sustainable, and why?*" Model: yours — CM/CAC ratio of 2.75 vs. 1.14.

Unit 3: Honest stories that sell

MC1. *"You ran a 3-day A/B test on two posters. Poster A: 8 customers stopped. Poster B: 6 customers stopped. The result is..."*

A. A clear winner — A is 33% better · B. Direction-suggesting, but the sample is too small for certainty — could be noise · C. A tie · D. Failed test

Answer: B · **Standard:** *CCSS-Math 7.SP.A.1 / 7.SP.A.2 — Use random sampling to draw inferences.* · **Strand outcome (CBM, 7):** *I can run a 3-day mini-marketing test (two posters, two messages) and report which performed better, by what measure, and what I'd do next.* · **Misconception targeted:** "*An A/B test result of 8 vs. 6 is a clear winner*" — corrects.

MC2. *"For a hypothetical product, you considered the claim 'EVERYONE LOVES IT.' This claim fails the 4-point Fair-to-Customer Check because..."*

A. It's not specific or verifiable · B. It's bandwagon — but bandwagon is allowed in ads · C. It's too modest · D. The font is too big

Answer: A · **Standard:** *CEE Standard 2 (Buying Goods and Services), Grade 8 — People can use information from sellers and outside reviewers to compare products.* · **Strand outcome (ESL, 7):** *I can apply a 4-point "fair to the customer" check (clear claim, honest price, real reviews, easy returns) to a real or fictional product page.* · **Misconception targeted:** "*Honest claims sell less than clever ones*" — corrects.

MC3. *"A customer says 'I'd buy that.' Is this evidence they will buy?"*

A. Yes, words are commitment · B. No, words are not behavior — only money down or a deposit is evidence · C. Yes, if 5 people say it · D. Only if they nod

Answer: B · **Standard:** *CASEL Social Awareness — Perspective-taking.* · **Strand outcome (CBM, 7):** *I can build a one-page "customer persona."* ·

Misconception targeted: "*If a customer says 'I'd buy it,' that's evidence*" — corrects.

SA1. *"For a hypothetical poster A/B test, Poster A got 16 stops in 3 days; Poster B got 11. Write a 3-sentence analysis: which performed better, by what measure, and what you'd do next."*

Model answer: "*Poster A had 45% more stops than B over 3 days, suggesting the 'soccer-day breakfast bar' headline outperformed the 'protein bar' headline. The sample is small — 27 stops total — so I'd repeat the test next week with a 5-day window for more confidence. If the gap holds, A becomes our default; if it reverses, the headline isn't the lever.*"

Scoring guide (3-2-1): - **3:** Names winner + measure + next step + sample-size hedge. - **2:** Winner + next step, no hedge. - **1:** Picks winner without hedging.

SA2. *"Run the 4-point Fair-to-Customer Check on a real ad you've seen: clear claim, honest price, real reviews, easy returns. Pass or fail each point."*

Model answer: "*Brand X protein bar ad. Clear claim: PASS — '15g protein, 200 calories.' Honest price: PASS — $2.49 visible. Real reviews: FAIL — testimonials are unattributed and no number of reviewers given. Easy returns: FAIL — no return policy mentioned. Two of four fail; I'd not recommend this brand to my persona.*"

Scoring guide (3-2-1): - **3:** All 4 points checked with specifics + a verdict. - **2:** All 4 checked, no verdict. - **1:** Fewer than 4 checked.

Performance Task: 3-Day Mini A/B Test (week-long, with daily logs).

Child runs a 3-day mini-marketing test on two posters or two messages, tracking customer-stops or sign-ups daily. Writes a 1-page report (winner, measure, sample-size note, next step) AND runs the 4-point Fair-to-Customer Check on the winning version. **Rubric:** Customer Research Rubric (Grade 7): (1) test setup fair (controlled variables), (2) data tracked daily, (3) interpretation hedges sample size, (4) Fair-to-Customer Check applied. **Strand outcome:** CBM + ESL.

Extension (advanced learner):

"*Your A/B test result reverses on Day 4 (B beats A). What does this teach you about the original 'A wins' conclusion? Write a 1-paragraph reflection.*" Targets epistemic humility; previews Grade 8 PDSA.

Unit 4: Feedback that fits

MC1. *"You designed a 3-question post-purchase survey. Which question is BEST for actionable feedback?"*

A. 'Did you like it?' · B. 'On a scale of 1-10 how was your experience?' · C. 'What's one specific thing that would have made this better?' · D. 'Would you recommend us?'

Answer: C · **Standard:** *CCSS-ELA W.7.7 — Conduct short research projects.* · **Strand outcome (OSO, 7):** *I can design a 1-week customer feedback system (a 3-question survey + a 1-question post-purchase prompt) and report what I learned.* · **Misconception targeted:** "*Vague survey questions yield useful data*" — corrects.

MC2. *"For your Day 1 customer assumption, you wrote 'soccer players want a sweet bar.' By Day 30, evidence shows they want a savory option. Your revision should..."*

A. Hide the original assumption · B. Document what changed AND why, in 3 specific revisions · C. Pick the new option without explanation · D. Stick with sweet, evidence be damned

Answer: B · **Standard:** *NGSS-ETS MS-ETS1-3.* · **Strand outcome (LRCI, 7):** *I can compare my "Day 1 customer assumptions" to "Day 30 customer evidence" and revise my product/messaging in 3 specific ways with reasons.* · **Misconception targeted:** "*Revising means I was wrong, which is bad*" — corrects.

MC3. *"Leading a 10-min team huddle, your highest-leverage move is..."*

A. Re-explaining the goal · B. Assigning one specific customer-research task to each teammate AND following up next day · C. Letting people 'check in' freely · D. Reading status reports

Answer: B · **Standard:** *CCSS-ELA SL.7.1 — Engage effectively in collaborative discussions.* · **Strand outcome (LTO, 7):** *I can lead a 10-minute team huddle in which I assign one customer-research task to each teammate and follow up the next day.* · **Misconception targeted:** "*Meetings are for talking*" — corrects.

SA1. *"Design a 3-question survey + a 1-question post-purchase prompt for your venture. Each question must be specific and actionable."*

Model answer: "*Survey: (1) 'What's one thing you'd want changed about your bar?' (2) 'Where would you usually buy this — game day, school, vending machine?' (3) 'Price you'd pay: $1, $2, $3?' Post-purchase: 'On a scale of 1-5, how likely are you to buy this again, and why?'*"

Scoring guide (3-2-1): - **3:** All 4 questions specific AND actionable (each maps to a possible change). - **2:** 3 of 4 specific. - **1:** 2 vague.

SA2. *"Document 3 revisions between your Day 1 assumption and Day 30 evidence. For each: state the original, the evidence that changed it, and the revised version."*

Model answer: "*1. Day 1: 'sweet bar.' Evidence: 4 of 5 soccer-mom interviews said savory. Day 30: 'savory bar.' 2. Day 1: '$3 price.' Evidence: 3 of 5 said $1.50 max. Day 30: '$1.75.' 3. Day 1: 'school cafeteria distribution.' Evidence: cafeteria has a 6-week approval cycle. Day 30: 'team group-chat pre-orders.'*"

Scoring guide (3-2-1): - **3:** All 3 revisions with original + evidence + revised version. - **2:** All 3 with revisions but evidence vague. - **1:** Fewer than 3 revisions.

Performance Task: 1-Week Feedback System Run (90 min planning + 1 week deployment).

Child deploys their 3-question survey + post-purchase prompt across a 1-week feedback cycle on a real (or simulated) classmate-customer base. Reports on what they learned AND makes 3 specific revisions to their product or messaging. **Rubric:** Customer Research Rubric (Grade 7): (1) survey deployed and at least 5 responses collected, (2) findings summarized in customer language, (3) 3 specific revisions documented with evidence. **Strand outcome:** OSO + LRCI + LTO.

Extension (advanced learner):

"*Two pieces of customer feedback contradict each other: 'less sweet' and 'more sweet.' How do you decide whose feedback to weight more? (Hint: consider buying behavior, not stated preference.)*" Anchors the "behavior > words" principle; previews Grade 9 customer discovery.

Total this grade: 12 MC + 8 SA + 4 Performance Tasks + 4 Extensions = 28 items.